GENERAL SPECIFICATION G3.41

Material Specification for Sewer Pipes

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G3.41 MATERIAL SPECIFICATION FOR SEWER PIPES

G3.41.1 Identification and Supply

(a) The Contractor shall ensure that all materials and workmanship follow Clause 10 of the Standard Conditions of Contract.

(b) Every pipe made shall be clearly and indelibly marked as follows:

(i) Type/shape of pipe - ‘Lead’, ‘Ordinary’ or ‘Inter-jack’;

(ii) Internal diameter of pipe / nominal bore size;

(iii) Date of manufacture;

(iv) Manufacturer’s name or mark;

(v) Manufacturing standard reference; and

(vi) Crushing strength in kN/m (open trench pipes) and maximum allowable jacking load in kN (jacking pipes).

(c) Pipe manufacture shall be arranged to permit the full completion of batching of pipe before testing. After the satisfactory completion of testing and approval of the pipes by the S.O. the pipes shall be stored during the period awaiting delivery. All pipes shall be fitted with an approved packing material such as Medium Density Fibreboard (MDF) at least 18mm thick.

(d) Full Quality Assurance records are to be provided to the S.O. for each pipe test and for each individual pipe, including, the date manufactured, date cleared after testing, date supplied to site, and maximum allowable jacking load. The S.O. will have the right at all reasonable times to enter the premises on which the pipes are to be manufactured, cured or stacked, to inspect and/or take samples.

(e) The manufacturer shall declare the factor of safety that relates to the design jacking load. The Contractor shall be responsible for calculating the allowable jacking load based on the manufacturers factor of safety, manufacturer’s maximum jacking load, soil information and method of jacking where lubrication may be utilized.

(f) Methods and working practices used to transport, stack, handle and install pipes, segments, rings and platforms shall comply with the manufacturer’s requirements.

G3.41.2 Supply and Laying of Pipes and Fittings

(a) The Contractor is required to schedule his own requisition and he shall ensure consistent delivery of pipes and fittings.

(b) All arrangements for the timely delivery of pipes and fittings to meet the progress of the work shall be solely the responsibility of the Contractor.

(c) The Contractor is required to plan and arrange the Works to ensure that standard pipe lengths are used between manholes including the short length pipes used next to the manholes. Where a pipe of odd length is, in the opinion of the S.O., required, the Contractor shall arrange for the casting of such odd, make up length of pipe or cutting of pipe when necessary. The Contractor shall allow in his rates for all such casting, cutting and waste.

(d) The Contractor shall provide a certificate to confirm the maximum allowable jacking load. The Contractor shall be responsible for calculating the maximum allowable in-situ jacking loads based on the soil information provided in the tender documents and shall also ensure that these loads are not exceeded.

G3.41.3 Cast Iron Manhole Frames and Covers

(a) The Contractor shall submit the following details of manufacture:

(i) Process of manufacture;

(ii) Details of manufacture, country or place of manufacture; and

(iii) Composition of materials.

(b) The manhole frames and covers to be installed under this Contract shall be sound and of the best quality. They shall conform generally to SS30: 1981 (Revised) or accepted equivalent standard.

(c) The manhole frames and covers shall be manufactured in accordance with the latest sewerage standard drawings available at PUB website.

(d) The manhole frames and covers shall be tested in accordance with SS30:1981 (Revised). The testing shall be conducted by a SAC-SINGLAS accredited laboratory. The test loads shall be as follows:

	Grade	Test Load (kN)
(a)	Heavy Duty	230
(b)	Medium Duty	75

(e) The load test shall be carried out on one (1) out of every batch of fifty (50) or less sets of manholes frames and covers.

(f) The Contractor must submit the Test Certificate for every batch of fifty (50) or less sets of manholes frames and covers to the S.O. to certify that the manhole frames and covers he intends to use in the Contract comply with the SS 30 : 1981 (Revised). Where the products are obtained from different batches, they must be accompanied by Test Certificates for the respective batches. Materials not accompanied by Test Certificate(s) shall not be used in the Contract.

(g) Each manhole cover must be imprinted with a serial number. The serial number imprinted on the manhole cover must tally with that shown in the Test Certificate accompanying it.

(h) The S.O. reserves the right, notwithstanding the aforesaid, to instruct the Contractor in writing to submit samples for load testing, if in his sole opinion, the quality of the materials supplied does not meet with the requirements. Upon receipt of written instruction from the S.O., the Contractor shall within seven (7) days from that date arrange for the load test to be carried out.

(i) If the sample selected fails the load test, the entire batch from which the sample is selected shall be rejected and the cost incurred in carrying out such test shall be borne by the Contractor. However, the S.O. may consider permitting a 100% test to be carried out on the whole batch from which the selected sample had failed the load test. Material from the batch, which passes the load test shall be accepted and those that fail shall be rejected, provided a written request for such test is made to the S.O.. All additional costs incurred for carrying out such test shall be borne by the Contractor.

(j) If the sample selected passes the load test, the cost incurred in carrying out the test will be borne by the Board. The cost shall be the cost charged by SPRING Singapore for conducting the test.

(k) The materials to be supplied under this Contract shall be sound and free from physical defects. All materials delivered shall be subjected to physical examinations; and any material which in the sole opinion of the S.O. or his representative is considered unsound or contains physical defects shall be rejected.

G3.41.4 Manhole Chambers – Precast Reinforced Concrete Pipes

(a) General

(i) The precast reinforced concrete chamber rings shall be manufactured in accordance with Drawing No. PUB/WRN/STD/003.

(ii) The materials and workmanship shall be of the best quality. The chamber rings shall meet the testing requirements specified.

(iii) The units shall be of reinforced Portland Cement Concrete with a minimum strength of C35/45. Each shaft ring shall be of uniform diameter and thickness of wall throughout its entire length and all the units shall be to the thickness, rebate length and dimensions given on the drawing.

(iv) The rings shall be manufactured by an approved centrifugal process, or vertical casting method.

(v) The reinforcement shall consist of hoops or spirals and longitudinal firmly secured together and the steel grid or grids so forming the reinforcement shall be centrally and accurately located as shown in the drawing by some positive method to maintain position during manufacture.

(vi) The units shall be designed to withstand the specified tests and any unit which in the sole opinion of the S.O., does not comply with the requirements of such tests, shall be rejected. The moulds and method of manufacture shall be such that the form and dimensions of the finished work are accurate within the limits specified, the surface and edges clean and true, the ends square with the longitudinal axis, and the concrete dense and homogeneous.

(b) Variation in Thickness of Walls

(i) Subject to the mean thickness of the wall of the rings being not less than that shown on the drawing, the radial thickness of the wall shall nowhere vary more than three (3) millimetres for the chamber rings. The mean thickness of the wall shall be ascertained by adding the measured least thickness to the measured greatest thickness and dividing the sum by two.

(c) Straightness

(i) The barrels of all rings shall be truly straight when tested on the inside with a rigid straight edge for their full length and on a line parallel to the longitudinal axis of the pipe.

(d) Joints

(i) All the units are to be formed with ogee joints in accordance with the drawing and all the joints are to be accurately and neatly formed so that all units fit neatly together. The ogee ends of each unit shall be parallel and square with the longitudinal axis.

(e) Lifting Holes

(i) Each unit shall have two lifting holes 40mm diameter formed through the concrete to facilitate handling and positioning of the units. The holes shall be carefully positioned clear of reinforcement and shall be left rough internally.

(f) Testing of Materials

(i) Testing of materials shall be conducted by a SAC-SINGLAS accredited laboratory.

(g) Testing Requirements

(i) Reinforced concrete pipe sections shall comply with the Singapore Standard SS 183. In addition, Crushing Test shall also comply with Clauses 23.2 and 23.3 of Singapore Standard SS 183 for Class 'L' pipe and the Water Absorption test shall comply with the Clause 15.1 of BS 5911-2:1982.

(ii) The selection of samples for testing shall be as follows:

1) Absorption Test: 1 per batch of 100 or part thereof

2) External Loading Test:

- Cracking Load 1 per batch of 50 or part thereof
- Ultimate Load 1 per batch of 100 or part thereof

(iii) Should the section selected for Absorption Test fail, two further samples from the same batch may be selected for testing and if both comply with the test the batch represented shall be accepted, but if either one of the samples fails the whole of the batch represented may be rejected.

(iv) Should the sections selected for the External Loading Test pass the test, the whole of the pipes comprising the batch shall be deemed to comply with this test. If the sample fails, two further numbers from the same batch may be selected for testing and if both samples comply with the test, the whole of the pipes comprising the batch shall be deemed to comply with this test. If any of the samples for retest fail, then the whole of the batch represented may be rejected.

(v) The Contractor must produce a SAC-SINGLAS accredited laboratory test certificate to the S.O. to certify that the precast concrete chamber rings he intends to use in the Contract have been tested and that the products comply with the relevant Standards before they can be accepted. Where the products are obtained from different batches, they must be accompanied by Test Certificates for the respective batches. Materials not accompanied by Test Certificate(s) shall not be used in the Contract. Every chamber ring must be stencilled with a serial number which must tally with that shown in the Test Certificate accompanying it.

(h) Marking of Rings

(i) The following information shall be clearly stencilled on the exterior surface of each section at the time of manufacture:

1) The internal diameters of the ring and on each ring the word TOP at the upper end;

2) Standard Reference;

3) Date of Manufacture;

4) The serial number in the order of manufacture; and

5) Manufacturer's name and registered trade mark.

(i) Early usage of precast chambers unless supplied under this Specification, the otherwise manufactured chambers are not allowed to be used for construction before 5 weeks after the date of casting.

(j) Physical Inspection

(i) The materials to be supplied under this Contract shall be sound and free from physical defects. All materials delivered shall be subject to physical examination; and any material which in the opinion of the S.O. or his representative is considered unsound or contain physical defects shall be rejected.

G3.41.5 Concrete Manhole / Sewer Pipe – Internal Protective Lining

(a) All exposed surfaces in concrete manholes/ chambers shall be lined with 2.5mm of approved HDPE lining. This shall include the underside of all intermediate concrete platforms, and roof slabs. To prevent slips and falls the HDPE lining is not to be applied to the top of intermediate platforms or roof slabs. The linings must be designed and formed with mechanical keys for keying into the concrete.

(b) Ends of the lining shall be embedded 50mm inside the concrete or sealed by approved HDPE angle fillets and welded in accordance with the manufacturer's instructions.

(c) The Contractor shall take all necessary measures to protect the HDPE lining from being damaged during installation and shall make good any damage arising thereof. The S.O. may order a test be carried out to detect any holes in the HDPE lining.

(d) Drawings and details indicating method of manufacture, lining, welding of joints, etc., together with test certificates of physical, chemical and biological properties of the HDPE shall be submitted for S.O.'s approval prior to manufacturing of any pipe. All tests shall be carried out by SINGLAS accredited laboratories and witnessed by the S.O. or S.O.'s representative.

(e) Reinforced concrete pipes of diameter 900 mm and above are to be lined with HDPE of minimum thickness of 2.5 mm. The lining shall be fixed to concrete surface by mechanical keys and cover not less than 330^0 of the internal circumference of the pipes. Alternatively, the lining may cover 360^0, provided holes are catered for at the trough of the pipe to allow releasing of water-pressure between the lining and the inner face of the pipe.

(f) The internal surface of the pipe shall be free from wrinkles, bulges, corrugations, burnt surfaces and imperfections. The lining shall be firmly keyed into the concrete with the back face of the lining flush against the concrete surface. The keys shall be continuous and capable of withstanding a back pressure of not less than 0.48 N/mm^2. Partially embedded keys will not be acceptable. There must be complete bonding in all welds.

(g) Only pipes manufactured and lined to the satisfaction of the S.O. are acceptable. Minor defects may be repaired with approval of the S.O. but each defect must be made good to the complete satisfaction of the S.O. who reserves the right to reject any pipe (or pipes) which in his opinion do not comply with this Specification and the required standard. In the event of the HDPE lining being damaged during the construction of the sewer, the whole pipe shall be replaced entirely at the Contractor's expense.

(h) When pipes are laid, care shall be taken to keep the cover flaps clear of the adjoining pipe socket and to place the pipe in the correct position so that the HDPE sheeting is evenly divided by the vertical pipe centre line.

(i) At regular intervals, the Contractor shall arrange for his pipe supplier to visit the site and weld up and test the HDPE joint flaps and end cover fillets. Not more than three (3) pipes shall be laid before the joints are welded up.

(j) The Contractor shall be required to afford all necessary assistance in keeping the pipes clear of mud and water during the jointing of the HDPE flaps.

(k) The Contractor shall be required to arrange the supply of water and power and for the proper ventilation of the pipeline during cleaning, tacking and welding and testing of the HDPE joints.

(l) On completion of the laying, jointing and welding, the Contractor shall be required to carry out a detailed inspection and test of the HDPE lining and joint welding in the presence and to the satisfaction of the S.O..

(m) HDPE linings to reinforced concrete manholes and pipes shall comply with the properties listed in Tables below.

(n) Physical Properties

Property	Allowable Standard	Method	Testing Frequency (minimum)
Density of base material	0.936 – 0.942 g/cm^3	ASTM D 792 / ASTM D 1505	90, 000 kg
Density of finished product	0.942 – 0.950 g/cm^3	ASTM D 792 / ASTM D 1505	90, 000 kg
Carbon Black Content	2 – 3 %	ASTM D 1603	9, 000 kg
Tear Resistance	125 N/mm	ASTM D 1004	20, 000 kg
Puncture Resistance	320 N/mm	ASTM D 4833	20, 000 kg
Elongation at yield	Longitudinal : minimum 12 % Transverse : minimum 12 %	ASTM D 6693 Type 4	9, 000 kg
Elongation at break	Longitudinal : minimum 700% Transverse : minimum 700 %		
Tensile strength at yield	Minimum 15 N/mm^2	ASTM D 6693 Type 4	9, 000 kg
Tensile strength at break	Minimum 27 N/mm^2		
Oxidative Induction Time (OIT) Standard OIT High Pressure OIT	 100 min 400 min	 ASTM D 3895 ASTM D 5885	90, 000 kg
Water absorption	0.085%	ASTM	
Water soluble matter	Zero	D 570	
Porosity	No pin holes	Spark Tester 7kV	
Coefficient of Linear Expansion	1.2 x 10^{-4} C^{-1}	ASTM E 831	
Plasticiser	Nil		

(o) Chemical Properties

Chemical Agents		Test Method	Change in Weight Not more than
Sodium Hypo-Chloride	1%		0.09%
Ferric Chloride	1%		0.09%
Sodium Chloride	5%		0.09%
Sulphuric Acid	20%	ASTM	0.09%
Nitric Acid	1%	D 543	0.09%
Sodium Hydroxide	5%	(7 days at	0.09%
Ammonium Hydroxide	5%	20ºC)	0.09%
Soap & Detergent solution	2%		0.09%

G3.41.6 Concrete Manhole / Tunnel Preformed Cover Fillets

(a) The Contractor shall include the supply and welding of pre-formed cover fillets or angles to seal the end of the HDPE lining at the manholes. The fillet or angle shall be of the same material and thickness as the HDPE lining of the pipes. The details of finishing, welding and fixing to the concrete walls adjoining the pipe end; spigot or socket shall be provided by the manufacturer. The Contractor shall be required to form any chase, rebate, fillet or recess required for the proper fixing and sealing of the cover fillets.

G3.41.7 Jacking Pipes

(a) Jacking Pipes - General

(i) The Contractor shall submit full details of his proposals for the pipes, giving detailed drawings showing sizes, reinforcement and type of joints, calculations, together with the name of the proposed manufacturer, the place of manufacture, and the manufacturing processes to the S.O. for approval. All workmanship and materials used in the manufacture shall be subject to the approval of the S.O. who shall be from time to time permitted to inspect materials at source and the manufacturing processes in the factory.

(ii) The pipes shall be sufficiently matured with adequate compressive strength before they are used in the construction of the Works. They shall be handled with extreme care to prevent the edges of the pipes from chipping. Repaired pipes shall not be allowed for use in the Contract. The S.O. may reject any pipes he considers not suitable for the Works and these rejected pipes shall be removed from the site immediately.

(iii) After factory testing and before despatch, every pipe and special shall be marked in accordance with the Standard used. In addition, each pipe shall be marked with a number corresponding with the order of manufacture. Test certificates from the manufacturers or other relevant authority shall be submitted to the S.O., prior to any pipes being used on site. A copy of the test certificates is to be retained by the Contractor on site and made available upon request from the S.O.. The test certificates shall confirm the allowable distributed and deflected jacking loads.

(iv) Before any particular pipe-jack length commences, sufficient pipes and, if required, intermediates jacking station assemblies shall be available to ensure continuous operation.

(v) In soft ground (i.e. SPT<4) where buoyancy or steering is difficult, the Contractor may use a Buoyancy Can on the rear of the Shield. In this instance, the Contractor may also physically connect the tunnelling shield to the first 2 or 3 jacking pipes.

(vi) Either the collar or the spigot of the jacking pipe shall incorporate an appropriate gasket or water seal/barrier within its fixed length.

(vii) The Contractor's manufacturer of jacking pipes shall be required to demonstrate a third party certified quality assurance and control programme, and comply with Eurocode/Singapore Standard regarding materials, mixing and placing, curing and storage of constituents. The manufacturer's premises and methods shall be open to inspection by the S.O. for the purpose of checking the quality of manufacture. The Contractor shall ensure that all necessary assistance is provided to the S.O. on each visit.

(viii) In the event that a pipe fails under load tests, the Contractor will present two additional pipes from the same batch for testing. If either test fails under testing then the consignment of pipes manufactured from the same batch shall be rejected. The future testing frequency shall also be increased following a failure.

(ix) The Contractor shall include the price for supply of pipes to be tested and no claim shall be entertained by the Board. This shall include consideration of the cost for testing the pipes.

(x) Packing material shall be resilient and shall distribute pipe stresses arising from jacking loads. The packing material dimensions and installation shall be approved by the S.O. prior to commencement of jacking operations.

(xi) Where required, provision shall be made for the injection of lubricating fluid or grout through pre-formed holes in the pipe walls. A minimum of three holes at equal circumferential spacing shall be formed through every pipe. Lubrication holes shall be fitted with non-return valves.

(xii) Lubrication or grouting holes shall not be provided in jacking pipes with internal diameters less than 900mm.

(xiii) The Contractor shall provide a certificate to confirm the maximum allowable jacking load. The Contractor shall be responsible for calculating the maximum allowable in-situ jacking loads based on the information from the Soil Investigation Report and shall also ensure that these loads are not exceeded.

(xiv) The material used for joint packing shall be medium density fibreboard or chipboard, with a minimum thickness of 18 mm. Any alternative proposed material must be shown by independent testing to have equal or superior properties. The packing material shall extend around the full circumference of the pipe joint, but its outer edge shall not be less than 40mm from the outer face of the pipe at the joint.

(xv) Pipes shall have roundness such that the difference between the minimum and maximum external diameter does not exceed 1% of the specified nominal external diameter or 6 mm, whichever is the lesser.

(xvi) Pipes shall have an outer circumference that is within ± 1% of the nominal circumference or within ± 12 mm, whichever is the lesser.

(xvii) Pipe joints shall be designed to be watertight and comply with the leakage, testing and infiltration requirements. The Contractor shall be responsible for design and performance of pipe joints under all loading conditions including those loads induced by geotechnics, construction, testing, highway traffic, and routine maintenance and operation.

(xviii) The minimum concrete grade of jacking pipes shall be C35/45 minimum.

(b) Jacking Pipes – Stainless Steel Fixed Collar

(i) All jacking pipes, whether reinforced concrete, polymer concrete or vitrified clay, shall be fitted with stainless steel fixed collars. The collar shall be austenitic Stainless Steel of Grade 316 with minimum chrome content of 17%, minimum nickel content of 8% and of minimum thickness of 6mm welded into a completely circular hoop. The jointing collar shall be of dimension and thickness to be approved by the S.O..

(ii) Welding of stainless steel collars shall be full penetration butt welds, continuous across the full width of the collar. The filler materials used for welds must be compatible with the stainless steel collar material. After welding, all oxides shall be removed and the internal surface of the finished collar shall provide a good sealing surface in contact with the sealing gasket. All 'heat tint' shall be eliminated using back-up gases during welding or by chemical or mechanical removal on completion of welding. All welds shall be full, through wall welds to minimise the presence of crevices and all welds shall be deemed to be as strong as parent metal.

(iii) The collar/spigot joint shall have sufficient strength to provide an adequate seal under all temporary and permanent conditions and shall be securely anchored into the pipe by the pipe manufacturer. The collar shall be securely anchored and fixed to the jacking-pipe. A stainless steel collar shall be fixed by means of suitable Grade 316 stainless steel anchors welded to the collar but independent of any reinforcement. All welding of the collars and anchors shall be appropriate for the grade of stainless steel used and shall include all precautions to prevent corrosion of the materials and the welds.

(iv) The jacking collar shall be used at all times when a pipe is being jacked. This jacking collar will be sufficiently rigid to allow the jacking pressure to be distributed evenly around the wall of the pipe.

(v) A minimum design pull-out load of 25 tonnes shall be demonstrated in calculations for the collar by the Contractor.

G3.41.8 Reinforced Concrete Pipes

(a) Reinforced Concrete Pipes for Open-Cut Construction

(i) Reinforced concrete pipes shall be designed, manufactured and tested in accordance with the following specification:

1) Reinforced concrete pipes shall be manufactured by centrifugal or other equivalent process to be approved by the S.O.. Design, manufacture and factory testing of the pipes and special shall be to SS 183 or BS EN 1916:2002 or A.S. 1342 or other acceptable Standard. The clear cover of concrete over steel reinforcement shall not be less than 50mm.

2) The Contractor shall submit full details of his proposals for the pipes, giving detailed drawings showing sizes, reinforcement and type of joints, calculations, together with the name of the proposed manufacturer, the place of manufacture, and the manufacturing processes to the S.O. for approval. All workmanship and materials used in the manufacture shall be subject to the approval of the S.O. who shall be from time to time be permitted to inspect materials at source and the manufacturing processes in the factory.

3) The pipes shall be sufficiently matured with adequate compressive strength before they are used in the construction of the Works. They shall be handled with extreme care to prevent the edges of the pipes from chipping. Repaired pipes shall not be allowed for use in the Contract. The S.O. may reject any pipes he considers not suitable for the Works and these rejected pipes shall be removed from the site immediately. After factory testing and before despatch, every pipe and special shall be marked in accordance with the Standard used. In addition, each pipe shall be marked with a number corresponding with the order of manufacture. Test certificates from the manufacturers or other relevant authority shall be submitted to the S.O..

4) Reinforced concrete pipe shall be designed such that the pipe is able to withstand the test loads in accordance with Clause 23.2.2 of SS 183. During the course of manufacture, the Contractor shall manufacture a pipe from the same concrete batch mix for the purpose of this load test. The permissible crack width shall not be more than 0.25mm when subject to Load Test Requirements under Clause 23.2.2 of SS 183. One such test shall be carried out to represent every 200 pipes.

5) The Contractor may additionally elect to manufacture two (2) Nos. of similar pipes at his own expense and from the same concrete batch mix provided his intention is made known to the S.O. who initiated the instruction.

6) In the event that the pipe fails the load test, the Contractor may present the two additional pipes manufactured for testing. If either fails the Test Requirements, the consignment of the Reinforced concrete pipes manufactured from the same concrete batch mix shall be rejected. The future testing frequency shall also be increased following a failure.

7) The Contractor, in his prices for the supply of pipes shall take into consideration the cost of the tested pipes and the testing, and no claim shall be entertained by the Board in respect of this.

(b) Reinforced Concrete Pipes for Trenchless Construction

(i) Reinforced concrete pipes supplied must comply with the provisions of SS 183 or BS EN 1916:2002. For the purpose of jacking operation, the reinforced concrete pipes must also comply with the following requirements:

1) All pipes except for those less than 900mm diameter shall be manufactured with at least one set of grout holes. Each set shall consist of three grout holes spaced at 120^0 on centres circumferentially located at the quarter points from either end of the pipe. Two of these grout holes shall be at the soffit of the pipe.

2) The precast reinforced concrete pipes shall be sufficiently reinforced with steel to withstand all stresses induced by handling, jacking, earth and water pressures and all working loads at the depths at which they are to be used without cracking, spalling or distortion. The pipes shall be of at least strength Class 'H'. A load factor of not larger than 1.5 shall be used in the calculations to determine the strength of the pipes required. The strength of the pipes shall be tested by the three edge bearing test. When subjected to the design load in such a test, the maximum crack width developed on the pipe shall not exceed 0.25mm. All such tests shall be carried out at the expense of the Contractor.

3) In addition, the reinforced concrete pipes must also be designed and endorsed by the Contractor's PE to have a minimum safety factor of two against cracking induced by the force during the jacking operation.

(c) Cement Grouting

(i) The Contractor shall be fully responsible for preventing the occurrence of voids outside the pipe and if they occur he shall fill them with cement grout at the earliest opportunity.

(ii) Immediately following the jacking operation the Contractor shall pressure grout the jacked section to fill all voids existing outside of the pipe. Grouting shall be from the interior of the pipe through grouting holes as specified.

(iii) Systems of standard pipe, fittings, hose, and special grouting outlets embedded in the pipe walls shall be provided by the Contractor. Care shall be taken to ensure that all parts of the system are maintained free from dirt. Grout composed of cement, sand and other approved compound and water shall be forced under pressure into the grouting connections at the invert and shall proceed until grout begins to flow from upper connections. Connections shall then be made to these holes and the operation continued to completion.

(iv) Apparatus for mixing and placing grout shall be of a type approved by the S.O. and shall be capable of mixing effectively and stirring the grout and then forcing it into the grout connections in a continuous uninterrupted flow.

(v) After grouting is completed, pressure shall be maintained by means of stopcocks or other suitable devices until the grout has set sufficiently. After the grout is set, grout holes shall be completely filled with dense concrete and finished neatly without evidence of voids or projections.

(vi) For pipes with HDPE linings, grouting shall be carried out and the Contractor shall ensure that the pipeline is watertight before proceeding with the jointing of the linings of the pipes and the patching of the lining over the filled grout holes.

G3.41.9 Vitrified Clay Jacking Pipes

(a) Vitrified clay pipes and fittings shall comply with either European Standard EN 295: 1991, Australian Standard AS 1741:1991 or other Equivalent Standard.

(b) All pipes shall be sampled and tested to the requirements of the relevant Standards. SPRING Singapore Test Report or SPRING Singapore Certificate of conformity for vitrified clay pipes certified under the SPRING Singapore Product Listing Scheme (PLS) is acceptable as proof of compliance with the required standards. Type-tested pipes by other local/ overseas test laboratories accredited with SAC-SINGLAS are also acceptable.

(c) All pipes and its joints must be watertight. Rubber sealing ring and other jointing materials used in the joint assemblies shall comply with the requirements as specified in BS or EN 295: 1991 or other equivalent standards.

(d) The chemical resistance, hydraulic roughness, abrasion resistance and shear resistance of all pipes and fittings shall be tested and comply with the requirements as specified in BS or EN 295: 1991 or other equivalent standards.

(e) Dimension Requirements

Vitrified clay pipes shall have minimum bore as follows:

Nominal Size (mm)	200	300	400	500	600	700	800	1000	1200
Minimum bore (mm)*	195	293	390	487	585	682	780	975	1170

* Note: Other nominal sizes greater than nominal size 200 mm may be manufactured to comply with this standard, providing that the minimum permissible bore is not more than 2.5% less than the nominal size, rounded to the nearest mm.

(f) Testing

(i) The S.O. may instruct the Contractor to send pipes and fittings selected from any batch of 50 pipes and fittings or less (if the number of pipes and fittings in each batch is less than 50) to a SAC-SINGLAS accredited laboratory for tests if in the S.O.'s opinion such test is necessary.

(ii) Should the results of the test fail to meet the specified strength, the full cost of the tests shall be borne by the Contractor and the whole batch of the pipes may be rejected by the S.O.. However, if the test results are satisfactory, the Contractor will be reimbursed for the cost of the test.

(g) Marking Requirements

(i) All pipes manufactured and supplied shall have the following marks indented on the exterior of the barrel:

1) The manufacturer’s name;

2) Date of manufacture;

3) Nominal bore size of pipe;

4) Crushing strength in kN/m; and

5) Standard reference.

(ii) All pipes and fittings shall be from only one (1) manufacturer. The Contractor shall forward the manufacturer's signed certificate stating that the pipes and fittings comply with the relevant standard and the results of all tests to the S.O..

(h) Vitrified Clay Jacking Pipes - Material and Joints

(i) The following sets out the requirements for flexibly jointed vitrified clay pipes for construction by trenchless installation technique including pipe jacking.

(ii) Vitrified clay pipes shall have minimum crushing strength as follows:

Nominal Size (mm)	200	300	400	500	600	700	800	1000	1200
Minimum Crushing Strength (kN/m)	48	72	96	120	144	168	192	240	288

(iii) All pipes or pipe sections shall be sampled after any grinding or cutting of ends and tested to the requirements of the relevant Standards.

(iv) When tested in accordance with EN 295-3:1991, the deviation from squareness measured at the pipe ends shall be not greater than 1 mm.

(v) The manufacturer shall declare the design jacking load and ensure the strength of the pipes must be more than 2 times the design jacking load.

(i) Jointing Sleeves

(i) The joints of the vitrified clay pipes shall be of the double spigot type that is suitable for direct jacking or micro tunnelling. The jointing sleeves shall be made of Type 316 stainless steel.

(j) Rubber Rings

(i) The rubber sealing rings and other jointing materials used in the joint assemblies shall comply with the requirements as specified in the relevant Standards.

(k) Packers/Buffer Rings

(i) The packers/buffer ring shall be made of standard flooring grade chipboard sheets.

G3.41.10 Vitrified Clay Pipes for Open Cut Construction

(a) Vitrified Clay Pipes for Open-Cut Construction – Material and Joint

The following sets out the strength requirements for flexibly jointed vitrified clay pipes for installation by open-cut method of construction. In-wall jointed (ogee) pipes shall not be permitted in the Works. Pipes shall be supplied by an approved supplier and shall bear the appropriate identification markings.

(i) Vitrified clay pipes shall have minimum crushing strength as follows:

Nominal Size (mm)	200	300	400	500	600	700	800	1000	1200
Minimum Crushing Strength (kN/m)	32	48	64	80	96	112	128	160	192

(ii) All pipes or pipe sections shall be sampled after any grinding or cutting of ends and tested to the requirements of the relevant Standards.

(iii) Pipes shall meet the tolerance for squareness at ends of pipes as specified in EN 295-1. This is fixed at 6mm for pipes up to and including nominal diameter 300mm and then at 2% of nominal diameter for larger sizes.

G3.41.11 Polymer Concrete Pipes

(a) Polymer Concrete Pipes for Open-Cut Construction

(i) All Polymer Concrete pipes are to be manufactured, cured and tested in accordance with DIN 54815, Parts 1 and 2, with minimum allowable compression strength of 80 N/mm2. A GRP former may be used for pipe manufacture.

(ii) All pipes shall be indelibly marked as outlined in DIN 54815, Part 1, Section 7 with the additional marking for lead “lead”, “Ordinary” or “inter-jack” pipes as required. This shall include DIN number, batch number, nominal diameter, manufacturer, date and location of manufacture.

(iii) Durability Requirements

1) All polymer concrete pipes shall be designed to withstand the required loading using pipe strength at the end of 100 years. The Contractor is responsible to submit the durability test reports to the S.O. for approval before he could proceed to order the pipe.

(iv) Polymer Concrete Pipes - Materials

1) The wall resin shall be polyester.

2) The coarse aggregate shall conform to a maximum grain size of 16 mm. The fine aggregate sand shall have a maximum grain size of 2mm, and the filler shall be quartzite powder. The aggregate, sand and powder shall be washed and dried prior to use.

3) The polymer concrete pipe shall comply with the chemical resistance requirements as outlined in Table 2.5(B). This includes chemicals with pH ranges from 1 to 10.

4) Gaskets for polymer pipes shall comply with the requirements of EN 681 or BS 2494. The testing of the gaskets or seals shall be undertaken by the pipe manufacturer.

5) Polymer concrete may be used to form the connection between shafts and tunnels.

(v) Polymer Concrete Pipes - Dimensional Tolerances

1) Polymer Concrete Pipe Dimensions shall comply with DN54815-1.

(vi) Joints

1) The polymer concrete pipes shall be jointed by Type 316 stainless steel sleeves with a permanently integrated double lip elastomeric seal, which meets the performance requirements of DIN 4060.

(vii) Polymer Concrete Pipes - Testing

1) Polymer Concrete pipes shall be tested for vertical crushing, bending, splitting, longitudinal compressive strength, and fatigue in accordance with DIN 54815 Part 2.

(viii) Joint Test

1) Joint test shall be in accordance with DIN 4060.

(ix) Crushing Strength Test

1) The crushing strength shall be tested in accordance with items 5.4.1 of DIN54815-2.

(x) Compressive Strength Test

1) The compressive strength shall be tested in accordance with the Test Method 5.4.3 of DIN 54815-2. The compressive strength shall be a minimum of 90 N/mm2.

(xi) Chemical Corrosion Test

1) The manufacturer shall submit bona-fide test results of the pipes corrosion resistance in compliance with the chemical resistance requirements as outlined in clause G3.41.5 (o) including chemicals with pH range from 1 to 10.

(xii) Durability Test

1) The manufacturer shall conduct a long term crushing strength test of pipe under media attack in order to show that the pipe can withstand the required loading using pipe strength at the end of 100 years. This test for 100 years loading under media attack is determined by extrapolating the results of a long-term test series with the regression analysis method A of EN 705, lasting at least 10,000 hours, using at least 18 test pieces. The inner surface of the test piece bottom is exposed to the test solution. For one test series, use a sulphuric acid solution of pH1. For another series, use a sodium hydroxide solution at pH10. The loading shall be so selected that the time to failure are distributed over the test duration. This test must be conducted by an independent test laboratory.

2) For testing purposes, production lot shall consist of all pipes having the same lot marking number but shall not exceed a total of 50 pipes. Pipe length, wall thickness, pipe and joint dimensions and three-edge bearing strength shall be verified by an independent testing laboratory and approved by the S.O..

(xiii) Polymer Concrete Pipe - Rejection of Material

1) Pipes with fractures or cracks passing through the barrel shall be rejected. Chips, fractures, or blisters on the pipe shall not exceed 5mm into the barrel. Repairs, if any, shall be made only with the approval of the S.O.. All pipes shall comply with the leakage test in accordance with DIN 4060.

(b) Polymer Concrete Pipes for Trenchless Construction

(i) Polymer concrete jacking pipes supplied must comply with the requirements of Polymer Concrete Pipes meant for open cut construction. For the purpose of jacking operation, the polymer concrete jacking pipes must also comply with the following requirements:

1) The Contractor shall provide a certificate to confirm the maximum allowable jacking load. The Contractor shall be responsible for calculating the maximum allowable in-situ jacking loads based on the soil information provided and shall also ensure that these loads are not exceeded.

2) The manufacturer shall declare the factor of safety that relates to the design jacking load.

(ii) Joints

1) Joints for polymer concrete jacking pipe shall consist of a seat, an elastomeric sealing element, a sleeve and a compression disc or cushioning ring.

(iii) Seat

The seat shall be either a formed or ground recess on the pipe ends.

1) Elastomeric Sealing Element.

2) The elastomeric sealing element is an elastomeric gasket whose configuration and location may vary with pipe diameter and will provide a watertight joint at allowed deflection angles and after compression from tunnelling operations.

(iv) Sleeve

1) The sleeve is an element which bridges between pipe sections. The sleeve shall be made of Type 316 stainless steel, which in conjunction with the sealing element forms joints, which meets the performance requirements of DIN 4060. The composite sleeve shall have corrosion resistance in continuous service with a pH between 1 and 10. Sleeves shall be either flush or smaller in outside diameter than the pipe.

(v) Compression Disc or Cushioning Ring

1) The compression disc or cushioning ring is a flat disc that conforms to the remaining ends of the pipe after the joint is formed. Its purpose is to distribute the jacking forces, which develop during pipeline installation. The width of the compression disc or cushioning ring shall not exceed the maximum wall thickness of the pipe at the joint, nor shall it extend into the flow line or inhibit the installation of the sleeve onto the spigot end of the connecting pipe. Compression discs or cushioning rings shall be factory fabricated and installed.

(vi) Grout Holes

1) The Contractor shall determine in his calculations of anticipated jacking loads, the need for bentonite or other lubricant to reduce jacking forces.

2) The Polymer Concrete Pipes shall be supplied with factory-installed grout connections of diameter, quantity and orientation compatible with the micro tunnelling Contractor's lubrication operation. As a minimum, one grout hole shall be installed on each pipe. Grout holes shall be fitted with countersunk full-face rubber gaskets to prevent infiltration of displaced earth during the jacking process.

G3.41.12 Unplasticised PVC Pipes for Drainlines and Sewers

(a) Unplasticised PVC Pipes for Drainlines and Sewers for Open-Cut Construction.

(b) Unplasticised PVC pipes below ground for drain lines and sewers shall comply with SS272:1983 including all current amendments.

G3.41.13 Gaskets

(a) Compression Gaskets - General

(i) Elastomeric gaskets shall be from approved manufacturers and, if required, shall be tested in the presence of the S.O.. The rubber gasket shall be nitrile butadiene or equivalent rubber compound suitable for use in the pipeline conveying sewage or industrial water.

(ii) The gasket shall not deteriorate with age, heat or exposure to air under normal storage conditions.

(iii) The test rig shall simulate a range of conditions of displacement and joint gap, including the worst combination to be encountered in the completed structure. In each test the water pressure shall be increased in increments of 0.5 % bar and held at each value for 5 minutes. The final pressure shall be at least 1 bar in excess of the maximum hydrostatic pressure to which the structure may be subjected. This pressure shall be maintained for 24 hours during which no leakage shall occur at the gasketed faces. Tests shall be carried out at normal ambient temperature.

(iv) The gasket shall function under all combinations of packing and displacement encountered in the completed structure.

(v) Prototype gaskets shall be fit-tested to assess stretch characteristics.

(vi) The material from which gaskets are to be manufactured shall withstand any aggressive response from the ground or ground water or, where the tunnel is to carry effluents or liquids, the medium contained in the tunnel. In particular the gasket material shall withstand chemical attack and biological degradation such that the gasket functions properly for the design life of the facility.

(b) Hydrophilic Gaskets-general

(i) Hydrophilic sealing material shall perform to the same effect as elastomeric gaskets. The composition and properties of the proposed material shall be agreed with the S.O. and sealing strips and joints shall be subjected to the same testing regime as set out for elastomeric gaskets.

(ii) Hydrophilic gasket material shall take into account the chemical composition of the groundwater.

(c) Gaskets for pipe-jack joints

(i) Gaskets for pipe-jack joints shall provide a seal against the ingress of groundwater during jacking and in the permanent condition. Gasket material shall comply with the requirements of BS 2494, including resistance to chemical attack and microbiological degradation. The joints supplied shall be of the Cornelius rubber ring type or similar approved and shall be capable of accommodating 20 deflection at each joint.

(ii) The gasket shall be lubricated with a product recommended by the manufacturer and agreed with the S.O..

G3.41.14 Cement Grout for Cavity Grouting

(a) Cement Grout for Cavity Grouting – Mix Design

(i) General purpose cement grout shall be mixed in accordance with the proportions given in the following Table. The water content shall be kept to the minimum required to ensure a smooth, fluid mix.

Class	Proportion by mass		
	Cement	**Sand**	**PFA**
G1	1	-	-
G2	1	3	-
G3	1	10	-
G4	1	-	10
G5	1	-	4
G6	1	-	½

(ii) Pulverised fuel ash shall not be used as a constituent of grouts that contain sulphate-resisting cement.

(iii) Grout shall be used within one hour of mixing.

(b) Special Grouts

(i) Where necessary due to the nature of the ground conditions or where adverse water conditions are anticipated then the requirements for the use of special grouts shall be stated in the Contract.

(ii) Details of accelerating and retarding agents for proposed inclusion within the grout mix shall be submitted to the S.O. for agreement. Any such proposal shall be submitted in conjunction with a statement that outlines the Contractor's interpretation of ground behaviour during tunnel construction.

(c) Mixing

(i) Grouts containing polymer additives shall only be mixed in a colloidal type mixer.

(ii) Special grouts from proprietary manufacturers shall be mixed and used in accordance with the manufacturer's instructions.

(d) Storage and Delivery

(i) Bagged grouts shall be stored under cover dry surroundings and on a suitable platform, clear of the ground.

(ii) Bulk deliveries of grout constituents shall be stored in appropriate silos with suitable dust control and batch weighing equipment.

G3.41.15 Liquefied Soil Stabilizer (LSS) for Filling of Abandoned Pipes

(a) Requirements for LSS

(i) The Contractor shall engage an approved LSS supplier in designing the LSS mix and shall be responsible to ensure that the LSS supplied fully comply with the requirements as tabulated below:

Characteristic	Gauge	Requirement
Strength	Unconfined Compressive Strength	≥ 100kPa (after 1 day curing) ≥ 490kPa (after 7 days curing)
Density	Minimum density	1.70 t/m^3 (after 7 days curing)
Bleeding	Maximum %	Less than or equal to 2%
Flowability	Diameter of Flow Test	Min 11cm (on site) Max 23cm (on site)
Setting	Time	Min 4 hours

(ii) In addition, the LSS mix shall possess both self-flowable and self-compactable characteristics, without compromising the above stipulated requirements, as aided/deliberate compaction to the fill is not possible along the entire underground pumping main.

(iii) The proposed grout shall be able to flow and fill up at least 150m length of existing pipe in one grouting operation.

(b) Placement for LSS

(i) The Contractor shall ensure that the grouting materials entirely fill the pipe and bleeding for air lock must be provided for this exercise.

(ii) Prior to the placement of LSS, the Contractor shall carry out flushing of the pipe and shall ensure the pipe is cleared of any stagnant water, debris or undesirable materials.

(c) Test Samples

(i) Samples of LSS for testing shall be taken from delivery trucks at the point of delivery, on site. Care shall be taken to obtain representative samples.

(ii) The test specimens shall be of diameter 80mm x 80mm height cylinders, cast in metal moulds. Each mould shall have a metal base plate having a smooth machined surface. The interior surfaces of the mould and base plate shall be slightly oiled before being filled.

(iii) The test cylinders shall be made by filling the mould completely without any compaction.

(iv) At least 4 specimens per 50m3 of LSS (or per pour of LSS should the pour is lesser than 50m3) shall be collected for Unconfined Compressive Strength (UCS) testing at approved laboratory. 2 each shall be tested for 3-day and 7-day UCS.

(v) The location of the LSS from which the samples are taken and the date shall be noted by a mark on the cylinder. Each shall be given a reference number and each individual sample shall be denoted by suffixes “a”, “b”, “c”, and “d” to that number.

(vi) The test specimens shall be stored at the Site, at a place free from vibration, under damp sacks for 24 hours, marked on top with the sample reference numbers as described above and buried in damp sand until the time for sending to the testing laboratory. They shall be well packed for transport in damp sand or other suitable damp material, and shall be similarly stored at the laboratory until the time of test. The specimens shall be kept at the Site as long as possible (at least for three-quarters of the period before testing).

G3.41.16 Packings

(a) Packing material shall be resilient and shall distribute pipe stresses arising from jacking loads. The packing material dimensions and installation shall be agreed with the S.O. prior to commencement of jacking operations.

G3.41.17 Pointing and Caulking Material

(a) Caulking

(i) Cementitious caulking compound cord shall be asbestos free.

(b) Pointing

(i) Cement for pointing shall comply with BS 12 or BS 4027 as appropriate, sand shall comply with BS 882 and be of a grading commensurate with the work. The mix shall be cement: sand 1:3 or otherwise agreed with water sufficient only to provide a workable consistency which can be rammed into the joint. Mortar shall be used within one hour of mixing.

G3.41.18 Approval of Materials

(a) All materials and equipment furnished by the Contractor shall be subjected to the inspection and approval of the S.O.. No materials shall be delivered to the site of Works without prior approval of the S.O..

(b) Where materials or articles are specified to comply with Standards, the Contractor shall supply the appropriate certificate of compliance, if required by the S.O..

(c) As soon as possible after the Contract has been awarded, the Contractor shall submit to the S.O., data relating to materials and equipment he proposes to furnish for the Works. Such data shall be of sufficient detail to enable the S.O. to identify the particular product and to form an opinion as to its conformity to the Specifications.

(d) Facilities and labour for the handling and inspection of all materials and equipment shall be furnished by the Contractor. If the S.O. requires, either prior to beginning or during the progress of the Works, the Contractor shall submit samples of materials for such special tests as may be necessary to demonstrate that they conform to the Specifications. Such samples shall be furnished, stored, packed, and shipped as directed at the Contractor's expense.

(e) The Contractor shall submit data and samples of material sufficiently early to permit consideration and approval before materials are necessary for incorporation in the Works; the quality of the samples representing the quality of the bulk of such materials which he is proposing to use. Any delay of approval resulting from the Contractor's failure to submit samples or data promptly shall not be used as a basis of a claim against the Board.

(f) In order to demonstrate the proficiency of workmen, the Contractor shall provide such samples of workmanship when requested by the S.O..

G3.41.19 Rejected Materials

(a) All materials that are not acceptable shall be removed from the site by the Contractor within seven (7) days after notification in writing by the S.O..

(b) Should the Contractor fail to remove the rejected materials from site within seven (7) days, the S.O. shall have the right to remove the materials and the cost incurred shall be a debt owing to the Board by the Contractor and this shall be deducted from any monies due or to become due to the Contractor.

END OF SECTION